Sensor Complementarity and Operating-Point Proxies in
Acoustic Emission and Passive Ultrasound Regression of the ISO 281 Viscosity Ratio

Frederik Appel Vardinghus-Nielsen

Department of Electrial and Computer Engeering, Aarhus University, Denmark
frederikavn@gmail.com

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Abstract

Inadequate lubrication causes an estimated 30–80% of rolling-element bearing failures, yet online, non-invasive lubrication condition monitoring (LCM) remains unsolved: lubrication-relevant signal content is weak and strongly confounded with operating conditions. This paper presents a controlled dual-channel study in which wideband acoustic emission (AE) and heterodyned passive ultrasound (US) are acquired simultaneously from a ball bearing across staircase speed sweeps (140–2977 rpm) and 13 temperature blocks (33–100 °C), spanning the boundary, mixed, and full-film regimes, with the ISO 281 viscosity ratio κ as a continuous, physics-grounded regression target. Because κ is determined by the operating point, the central methodological question is what a feature–κ regression actually senses: a two-stage decomposition shows that acoustically inferred operating points alone recover $R^2 = 0.826$ of the direct model's 0.834, while within-speed-step analysis isolates a genuine temperature-mediated lubrication channel (20–500 kHz mobility, $\rho = -0.56$ at fixed drive state). Evaluation uses visit-grouped data splits that remove the near-duplicate leakage of windowed acquisition; recorded content above the AE coupler's 1 MHz low-pass is excluded from analysis. Under this protocol, gradient boosting attains hold-out $R^2 = 0.834$ (AE) and 0.653 (US); fusing the channels raises performance to 0.908 (ΔRMSE = 0.0281, significant on all tests), answering the long-argued but previously untested question of AE–US complementarity. All results derive from repeated grouped cross-validation with corrected significance tests.

Contents

Todo list

Nomenclature

Symbols

κ	ISO 281 viscosity ratio, ν/ν_1
ν	kinematic viscosity at operating temperature [$\mathrm{mm^2\,s^{-1}}$]
ν_1	reference viscosity for full film formation [$\mathrm{mm^2\,s^{-1}}$]
n	rotational speed [rpm]
d_{pw}	mean pitch diameter [mm]
N	samples per sweep
x_n	voltage sample n
S_k, f_k	one-sided FFT magnitude and frequency at bin k
f_c	spectral centroid [Hz]
σ_w	magnitude-weighted spectral bandwidth [Hz]
ρ, r	Spearman and Pearson correlation coefficients
R, k	cross-validation repetitions and folds

Abbreviations

AE	acoustic emission
CV	cross-validation
EHL	elastohydrodynamic lubrication
EMI	electromagnetic interference
HO	hold-out (test set)
LCM	lubrication condition monitoring
MAE	mean absolute error
RMSE	root mean square error
SHAP	SHapley Additive exPlanations
US	passive ultrasound
VFD	variable-frequency drive
VIF	variance inflation factor

1 Introduction

Rolling element bearings are among the most prevalent components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—through starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately causes premature failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received far less attention than vibration-based fault detection and remaining-useful-life estimation, which are now mature and widely deployed [4, 5]. The lubrication signal is harder to exploit: its acoustic and vibratory signatures are weaker than the impulsive signatures of spalling or raceway cracks and are strongly entangled with speed and load, so operating-condition effects readily mask the lubrication-dependent component [5–7]. Within bearing fault diagnosis, this entanglement has motivated a substantial recent literature on diagnosis under varying and unseen working conditions [8–10]; for LCM the problem is sharper still, because the lubrication target itself is derived from the operating point—a confound this paper addresses with an explicit diagnostic decomposition (Section 5.4). Yet accurate online LCM would enable condition-based relubrication—applying lubricant when and in the quantity actually needed, rather than on fixed schedules that systematically under- and over-lubricate [11].

The dominant bearing-monitoring approach, vibration analysis, detects macroscopic faults such as raceway spalling and rolling-element damage through characteristic-frequency analysis. Classical characteristic-frequency analysis is, however, poorly matched to the subtle, non-stationary tribological changes of film formation and breakdown, which lack the stable periodic signatures of structural faults [5, 12]; recent data-driven work nonetheless recovers lubrication-state information from vibration, typically from high-frequency broadband energy rather than from characteristic frequencies [13]. Temperature monitoring and oil analysis add complementary information but are indirect, subject to sampling delay, or require access to the lubricant reservoir, and none provides a direct, continuous, non-invasive indicator of the instantaneous lubrication regime under operating conditions [11, 14].

Passive, non-invasive sensing, e.g. acoustic emission (AE) and passive ultrasound (US), is practically attractive for industrial LCM because it requires no bearing modification, no direct access to the lubricant, and can be retrofitted to bearings in service. AE captures the broadband stress waves emitted by asperity contact and microcrack events, while passive US underlies the threshold-based relubrication tools already in routine industrial use, in which the broadband ultrasonic level is compared against a baseline and relubrication is triggered once it rises by a set margin [1, 15, 16]. The physical origins of both signals are detailed in Section 2. Despite the practical maturity of threshold-based US and the demonstrated usefulness of AE, the two channels' signal-processing behaviour has not been characterised simultaneously within a single regression framework against a physics-grounded lubrication reference.

The sensitivity of AE to lubrication regime is well established. Miettinen et al. [17] showed that AE pulse count and RMS distinguish boundary, mixed, and full-film regimes and respond to base-oil viscosity and thickener content in grease-lubricated bearings. Yoshioka and Shimizu [18] combined AE and vibration monitoring on a grease-lubricated deep-groove ball bearing using envelope demodulation and statistical AE parameters. Tandon et al. [19] found AE and shock-pulse sensors to outperform accelerometers for detecting grease contamination in a ball bearing—the scenario closest to the present study. More recently, Cornel et al. [12] showed that AE detects lubrication-sensitive surface damage such as micro-pitting and smearing up to eight hours earlier than vibration monitoring, Wang et al. [20] developed AE fingerprint features with dynamic thresholds for bearing fault detection, and Hou and Zhang [21] derived amplitude- and energy-centre features that classify lubrication status across speeds and viscosity levels, though without a continuous regression target, and with no variation of temperature.

On the signal-processing side, Renhart et al. [22] applied octave filter-bank decomposition with RMS–friction features to characterise AE tribological events, and Marticorena and García Peyrano [6] showed that spectral correlation of AE resolves lubrication-regime changes under varying load, demonstrating that kinematic and acoustic features can partially decouple operating-condition confounds. The physical link between film adequacy and the AE signature has been formalised by Liu et al. [23] and Zhang et al. [7], whose coupled vibration–AE models tie emission intensity and frequency content to EHL film thickness and the rate of asperity contact. For the passive-ultrasound channel, Jakobsen et al. [1, 16] showed that broadband acoustic features carry measurable information about the ISO 281 viscosity ratio κ.

A recurring limitation of data-driven LCM is its reliance on discrete fault labels or qualitative lubrication states (e.g. [13]) rather than a physically interpretable, continuous target. The ISO 281 viscosity ratio κ removes this limitation: it is the ratio of the lubricant's actual kinematic viscosity at operating temperature to the minimum viscosity required for full EHL film formation at the current speed and bearing geometry, mapping the instantaneous operating point to a standardised lubrication-adequacy index (Section 2). A model trained to predict κ implicitly learns the joint dependence on speed, temperature, and lubricant properties, and its output is directly interpretable against established regime boundaries. Despite this, κ has seldom been used as a regression objective in acoustic condition monitoring.

The clearest precedent is the work of Jakobsen et al. [1, 16], who established κ regression for passive bearing acoustics, predicting κ from low-cost MEMS microphone and accelerometer features using LASSO and neural networks. The present study adopts their κ target and extends it in three respects: it adds a wideband AE channel, analysed up to the 1 MHz limit of the AE coupler's band-pass, acquired simultaneously with the passive US channel, extracts frequency sub-band features from both signals, and tests whether the two modalities are complementary. This last point is an open question: AE–US complementarity is argued on physical grounds in reviews [5, 24] but has not been quantified experimentally, and it carries direct practical weight, since a second sensor adds cost and complexity and is justified only if it supplies non-redundant information. No prior study has simultaneously acquired AE and passive US from a single ball bearing and quantitatively assessed their joint value for κ regression.

This study deliberately fixes the bearing, lubricant, and load, varying only rotational speed and temperature. Holding the mechanical configuration constant isolates the lubrication-dependent acoustic response from the bearing-to-bearing variability and degradation processes that would otherwise confound it—a prerequisite for establishing whether AE and passive US carry κ-relevant information at all. The resulting limits on generalisation across bearing types, lubricants, and load regimes are genuine, but not addressed here.

Contributions

This paper makes the following contributions:

1. **A confound-decomposition methodology for acoustic LCM.** Because κ is determined by the operating point, any feature–κ regression risks acting as an operating-point soft sensor. A two-stage decomposition (features $\rightarrow$ RPM/temperature $\rightarrow$ ISO 281 mapping, recovering $R^2 = 0.826$ of the direct model's 0.834) quantifies the proxy share, while within-RPM-step correlation analysis of 20–500 kHz mobility (median $\rho = -0.56$ with temperature at fixed drive state) isolates the genuine lubrication-channel sensitivity.

2. **The AE–US complementarity question is answered quantitatively.** This is the first study to simultaneously acquire wideband AE and heterodyned passive US from a single ball bearing and assess their joint value: fusing the channels reduces hold-out RMSE from 0.109 to 0.081 (ΔRMSE = 0.0281), significant under repeated grouped cross-validation and all hold-out tests, with the extended 20–100 kHz US band contributing selected features to the combined model. AE outperforms US on the hold-out set, while fold-level inference between the single channels is inconclusive under group-respecting cross-validation.

3. **A leakage-aware, instrumentation-honest evaluation protocol.** Recorded content above the 1 MHz low-pass of the AE coupler (Kistler Type 5125C)—present

in the digitised signal despite that hardware filter, and of undetermined origin—is excluded from analysis. Visit-grouped data splits remove the near-duplicate leakage inherent to windowed acquisition of slowly varying protocols. Under matched random-split protocols, performance equals prior κ-regression work; the stricter visit-grouped figures are, to the authors' knowledge, the first reported for this task.

4. **A dual-channel dataset and reproducible evaluation protocol.** 8 504 sweeps spanning the boundary through full-film regimes ($\kappa \in [0.04, 1.64]$) with an analytically derived ISO 281 κ reference, evaluated under repeated nested cross-validation (25 outer scores) with Nadeau–Bengio-corrected and Holm–Bonferroni-adjusted inference.

Align with final data availability decision (full deposit, features+code only, or on request).

The remainder of the paper is organised as follows. Section 2 reviews the theory of κ and the physical origins of AE and US signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal-processing and feature-extraction pipeline, including the characterisation of the frequency sub-bands. Section 5 reports the feature analysis and the operating-condition confound decomposition. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings, and Section 8 concludes.

The overall pipeline of the experiment, signal processing, modelling, and evaluation is shown in Figure 1.

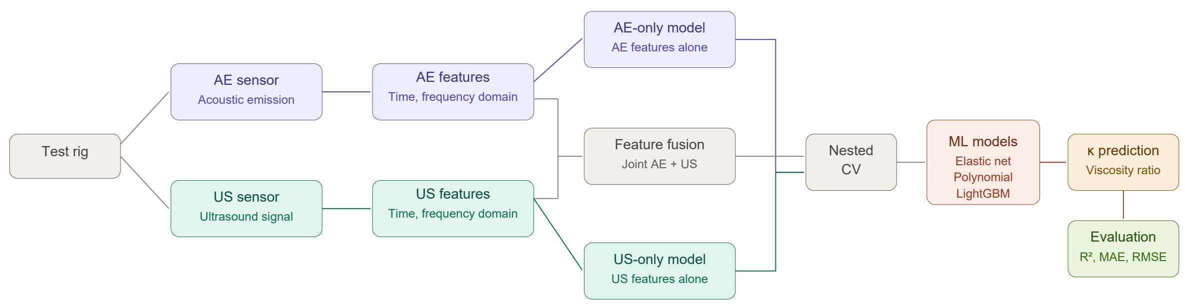

Figure 1: Overview of the experimental and modelling pipeline. AE and passive US signals are acquired simultaneously from the test rig and decomposed into broadband and sub-band features. Three sensor configurations (AE-only, AE+US fusion, US-only) are evaluated across three regression model families under repeated nested cross-validation, with the viscosity ratio κ as the regression target.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

Rolling element bearing contacts operate in one of three lubrication regimes depending on whether the lubricant film is sufficient to fully separate the rolling surfaces. In the full-film elastohydrodynamic (EHL) regime the surfaces are completely separated and asperity contact is suppressed; in the mixed regime the film is partially formed and intermittent asperity contact occurs; in the boundary regime the film is inadequate and metal-to-metal contact dominates. The ISO 281 standard characterises these regimes through the viscosity ratio κ, with boundaries at $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1$ (mixed), and $\kappa \geq 1$ (full film).

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu}{\nu_1(n,\, d_{pw})}, \tag{1}$$

where ν [$\mathrm{mm^2\,s^{-1}}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [25]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000\, n^{-0.83}\, d_{pw}^{-0.5} & n < 1000\,\mathrm{rpm}, \\ 4\,500\, n^{-0.5}\, d_{pw}^{-0.5} & n \geq 1000\,\mathrm{rpm}. \end{cases} \tag{2}$$

The operating viscosity ν is computed from the ASTM D341 Walther viscosity–temperature equation fitted to the lubricant's 40 °C and 100 °C reference values, with a standard value of $\lambda = 0.7$. Since ν_1 decreases with increasing rotational speed—faster rotation enhances hydrodynamic film entrainment, requiring less viscosity to achieve full EHL—κ captures the combined effect of speed, temperature, and lubricant grade on lubrication adequacy at a given operating point. Importantly, κ assumes nominally intact lubricant: the Walther equation reflects the properties of fresh grease, so κ does not encode viscosity changes arising from lubricant degradation, contamination, or starvation. Extending the regression framework to track such changes would require either continuous viscosity updates or an alternative target variable.

2.2 Physical origins of AE and passive ultrasound signals

Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings [17, 22]. Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing [12, 22]. The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime [12, 17]. Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors for bearings respond from tens of kHz into the MHz range and require correspondingly high sampling rates to capture the full frequency content.

Passive ultrasound sensing detects structure-borne acoustic energy above the audible range. As a modality it is not intrinsically narrowband—ultrasonic emissions span from roughly 20 kHz into the MHz range, overlapping much of the AE band—but the instruments most widely deployed for bearing monitoring operate in a narrow band around 38–40 kHz. In this band the sensor amplitude is modulated by the contact-stiffness and viscous-damping changes associated with lubricant film formation: as the film degrades and asperity contact increases, the ultrasonic level at the bearing housing rises [15]. Commercial instruments (UE Systems, Sonotec) exploit this by comparing the level against a reference baseline and triggering relubrication when a defined threshold is exceeded, making threshold-based passive US the most widely deployed non-invasive LCM method in routine industrial use. The passive US probe used in this study follows the same design: it senses around 38–40 kHz and heterodynes (down-mixes) that band to an audio-frequency baseband (Section 3). In practice, however, the recorded signal also contains energy in bands above the nominal baseband. As this higher-frequency content may nonetheless carry lubrication-relevant information, it is retained in the analysis rather than discarded, without attributing it to a specific physical origin. However, the level is also sensitive to rotational speed and load independently of lubrication condition, which limits the reliability of threshold-based approaches under varying operating conditions.

Check that this is correct

The two modalities thus respond to the same underlying phenomenon—increasing asperity contact as the lubricant film degrades—but through different physical mechanisms and at different frequency scales: AE captures individual high-frequency transient events ($\geq$50 kHz) from discrete micro-fractures and asperity collisions, while passive US, after the probe heterodynes its 38–40 kHz sensing band to an audio-frequency baseband, reflects the lower-frequency bulk acoustic envelope comprising structural resonances, friction noise, and the aggregate effect of contact activity, together with residual energy above the baseband whose information content is assessed empirically. Both channels therefore carry information about κ, and the question of whether they provide complementary or redundant lubrication-relevant information is addressed experimentally in Sections 3–6.

3 Experimental Setup

3.1 Test rig and bearing

Include picture of test rig

The test rig is a custom-built bearing test stand at Aarhus University (Herning, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter d_{pw} = 38.0 mm, 9 ceramic rolling elements, dynamic load rating 14000 N) is driven by a variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt) .

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz (nominal; the measured comb spacing is ≈4.7 kHz) and its multiples in the low-frequency range of both sensor signals. These are partially filtered out

3.2 Operating conditions

The experiment follows a protocol which varies the operating conditions through changes in rotational speed (rpm) and housing temperature (°C). This consists of a staircase sweep from 100 to 3000 RPM and back down to 100 RPM in steps of 100 RPM with a 60-second hold at each step. This staircase is repeated for each of 13 temperature blocks (40–100 °C in 5 °C increments) where temperature is controlled with . The block temperature set-point is changed between staircases rather than during the high-speed hold: after each downsweep the rig is held at the low-rpm bottom of the staircase (≈100 rpm) while the housing is brought to the next set-point and allowed to stabilise thermally. The recorded telemetry contains no true standstill (RPM = 0) sweeps; the only near-stationary data is a single warm-up segment (37–43 °C) acquired before the first staircase, which is excluded from analysis (Section 3.3). The hold at the 3000 rpm staircase peak is substantially longer than the ordinary 60-second steps (approximately 20 minutes), so this operating point is strongly over-represented in the data: it accounts for roughly a third of all sweeps. The applied radial load is a static dead-weight, held constant throughout the experiment at approximately 61 kg (≈ 600 N). The total duration of the experiment is approximately 14 hours. The measured operating conditions span 140–2977 rpm and 33–100 °C; the measured temperature range extends below the commanded 40 °C floor (down to ≈33 °C), which we tentatively attribute to the inherent undershoot and instability of the temperature control rather than to any commanded set-point.

The lubrication conditions in the bearing are not expected to change during the experiment in other ways than what is entailed by rpm and temperature and the effect of these upon the lubricant. It is not expected or assumed that any degradation, starvation or contamination of the lubricant happens.

The operating conditions are controlled through . The measured operating conditions space, distribution and time variation are seen in Figures 2 and 3. The κ value for each sweep is computed according to Equations (1) and (2) using the per-sweep measured RPM

What is the amount?

Do we need NLGI cosistency grade?

How is temperature controlled and measured?

Explain the process and expected outcome.

How is RPM controlled

With what?

The original protocol described a 2-min RPM = 0 standstill at each block start, and temperature, with ν evaluated via the ASTM D341 Walther viscosity–temperature equation.

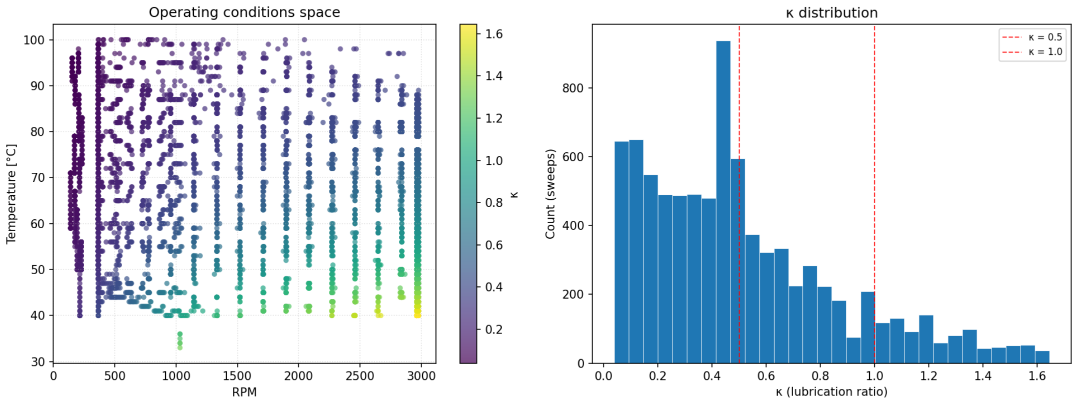

Figure 2: Left: space of the measured operating conditions (RPM and temperature) during the experiment. Each point corresponds to a 20ms window of sampling at a given RPM and temperature. The colour of each point corresponds to the derived κ value for that sweep, with warmer colours indicating higher κ values. Right: distribution of the derived κ values.

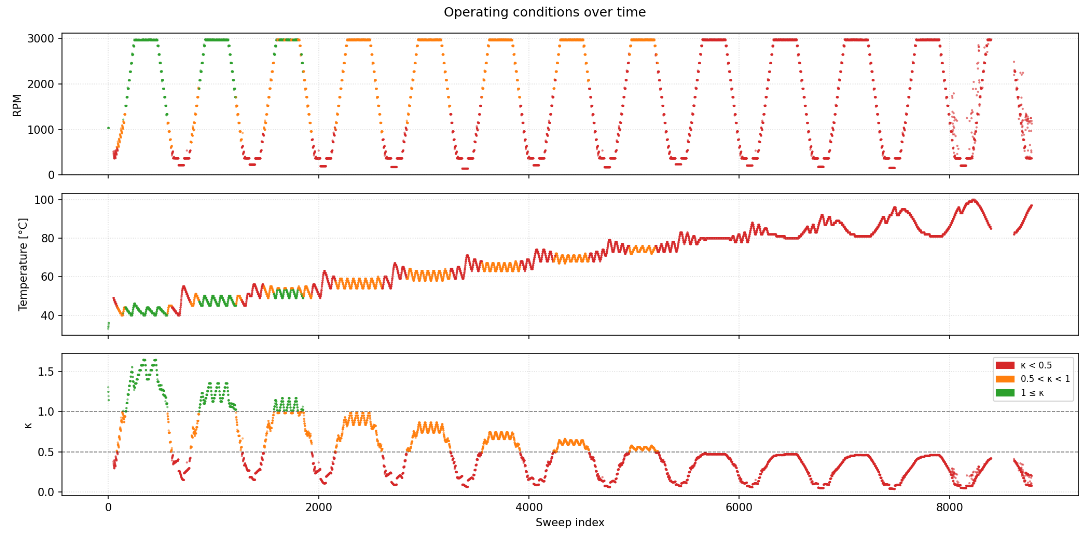

Figure 3: Measured operating conditions (RPM and temperature) for each sweep during the experiment. Each point corresponds to a 20ms window of sampling at a given RPM and temperature. The colour of each point corresponds to the derived κ value for that sweep, with warmer colours indicating higher κ values.

Consider inverting the colors. Green should be better, right?

3.3 Sensor configuration and data acquisition

Two sensors simultaneously acquire signals from the bearing housing through a shared oscilloscope which samples at a shared sample rate of 12.5 MHz:

Acoustic emission sensor: A Kistler Piezotron Type 8152C1 wideband piezoelectric AE sensor with integral impedance converter, bonded to the bearing housing. The sensor is ground-isolated, rated to 165 °C, and has a nominal ±10 dB frequency range of 100–900 kHz (sensitivity 48 $\mathrm{dB_{ref}}$ 1 V/(m/s)). Its output is conditioned by a Kistler Type 5125C AE Piezotron coupler, which powers the sensor and applies a Butterworth band-pass filter (50 kHz high-pass, 1 MHz low-pass) to the raw AE signal before it is digitised. This *coupler* band-pass, not the sensor element, sets the 1 MHz upper analysis limit used throughout the paper.

Ultrasound sensor:

Temperature sensor:

RPM sensor:

Load sensor:

AE sensor/coupler bullets written from the Kistler datasheets in `docs/`. Remaining bullets (US probe, temperature, RPM, load) await collaborator input. Temperature via an Omron controller?

The shared sample rate and DAQ setup ensures that AE and US signals share timestamps.

Acquisition windows (sweeps). Data are acquired as short windows rather than continuously: every ∼5.9 s the oscilloscope captures a 20 ms window ($N \approx 2.5 \times 10^5$ samples per channel) containing time-synchronous AE and US waveforms, together with the mean RPM and housing temperature over the window. Throughout this paper, one such window is referred to as a *sweep*, and a sweep is the unit of observation for all feature extraction, model training, and evaluation. After excluding an initial stationary warm-up segment recorded before the first staircase, the protocol yields 8 739 raw sweeps, of which 235 with measured RPM above the 3000 rpm protocol maximum are discarded, leaving 8 504 sweeps per channel (median 10 sweeps per staircase hold). The resulting κ distribution spans [0.04, 1.64] with mean 0.51 (Figure 2).

4 Signal Processing and Feature Extraction

4.1 Waveforms and spectrograms

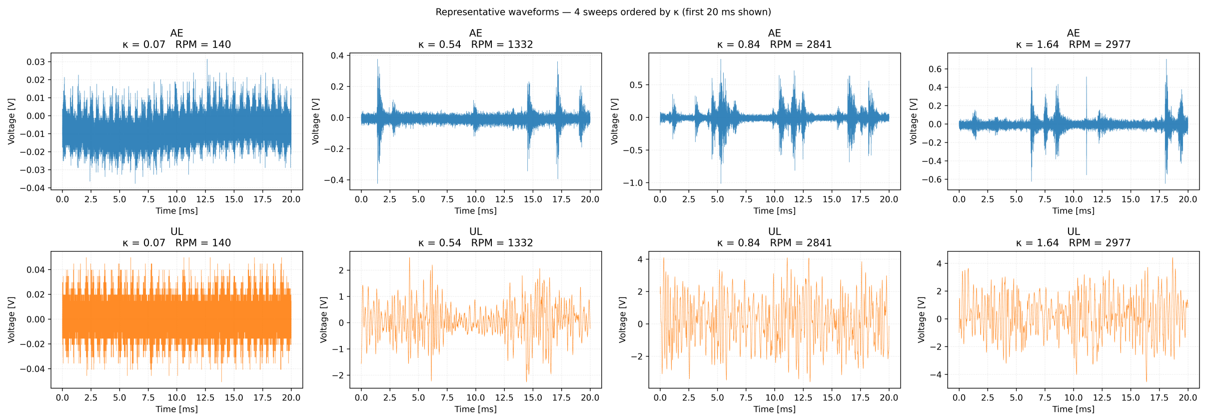

Figure 4: Waveforms of AE and US signals from the same sweeps for different values of κ. AE shows clear pulses of energy from asperity contacts which modulate the signal afterwards. US shows a more regular, lower frequency, non-stationary waveform.

Spectrograms of the raw signals, ordered chronologically by sweep, are shown in Figure 5 to illustrate the frequency content and its variation with operating conditions across the protocol.

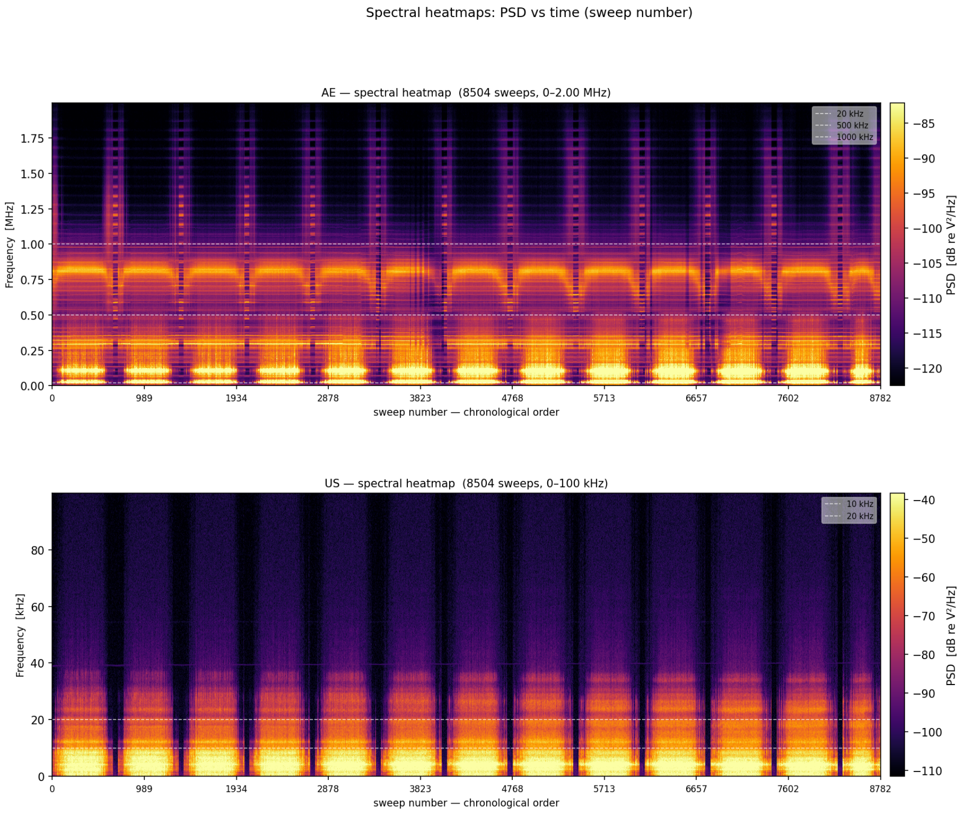

Figure 5: Spectrograms of the raw AE and US ordered by sweep number (chronological). The dashed lines show the sub-bands which are later used in feature generation.

4.2 Sub-band decomposition

The AE and US signals are decomposed into frequency sub-bands prior to feature extraction, since the information on lubrication conditions contained in the signals is expected to be localised in sub-bands rather than wholly contained in the broadband signals [7, 22]. Adaptive decompositions such as symplectic geometry mode decomposition offer data-driven alternatives for isolating informative components [26]; fixed, physically motivated bands are preferred here so that features remain directly comparable across sweeps and interpretable against the sensing physics. For the AE channel, two sub-bands are extracted in addition to the broadband signal: 20–500 kHz and 500 kHz–1 MHz which contain bands of spectral energy which upon visual inspection of Figure 5 vary with the operating conditions. Content above the 1 MHz coupler low-pass (Section 3.3) is excluded from the feature set. For the US channel, the signal spans the probe's 0–100 kHz acquisition band, and three sub-bands are extracted in addition to the broadband signal: 0–10 kHz and 10–20 kHz within the nominal heterodyned baseband (Section 2.2), and 20–100 kHz capturing the above-baseband content whose information value is assessed empirically. The sub-bands are summarised in Table 1. All decompositions use 5th-order Butterworth filters applied forward–backward (`sosfiltfilt`), giving zero phase distortion and a 10th-order effective magnitude response.

Table 1: Frequency sub-bands used for signal pre-filtering before feature extraction.

Channel	Sub-band	Filter type	Frequency range
AE	Broadband	–	0–1 MHz
AE	Sub-band 1	Band-pass	20–500 kHz
AE	Sub-band 2	Band-pass	500 kHz–1 MHz
US	Broadband	Low-pass	0–100 kHz
US	Sub-band 1	Band-pass	0–10 kHz
US	Sub-band 2	Band-pass	10–20 kHz
US	Sub-band 3	Band-pass	20–100 kHz

4.3 Sub-band signal validation

The retained sub-bands carry bearing-generated content rather than drive EMI. The VFD injects a switching comb at ≈4.7 kHz and its harmonics (Section 3.1); these lie below the 20 kHz lower edge of the retained AE bands and so cannot masquerade as bearing signal within them. More directly, total band power rises with rotational speed in both bands—$\rho = +0.91$ for 20–500 kHz (at 45–55 °C) and $\rho = +0.69$ for 500 kHz–1 MHz (Figure 6)—as expected for mechanically generated contact and rolling noise [7, 17], rather than the fixed-frequency behaviour of an electromagnetic-interference source.

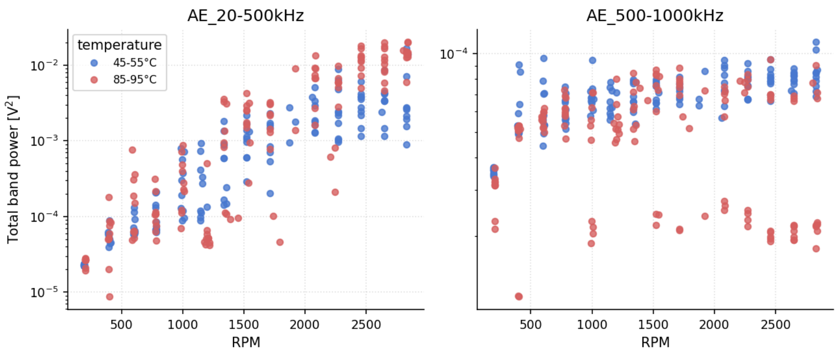

Figure 6: Total power of the two retained AE sub-bands as a function of RPM at fixed temperature windows. Both bands rise with speed, consistent with mechanically generated emission.

4.4 Feature extraction

Features are computed from the voltage signal of each sweep–sensor pair, for the broadband signal and for each sub-band. In total 14 features are computed, comprising 8 time-domain and 6 frequency-domain statistics, as listed in Tables 2 and 3. The set is deliberately canonical: commonly used companions such as standard deviation, variance, peak, impulse factor, and RMS frequency are exact transforms or products of the retained features (e.g. std = RMS for zero-mean band-filtered signals; peak = crest × RMS; impulse = crest × shape; and the RMS frequency satisfies $f_{\mathrm{RMS}}^2 = f_c^2 + \sigma_w^2$, where f_c

is the spectral centroid and σ_w the spectral bandwidth) and are therefore omitted, so that no candidate feature is a deterministic function of another. The frequency-centroid information that RMS frequency would carry is instead represented by Hjorth mobility, its time-domain counterpart: mobility $= \sqrt{\operatorname{Var}(x')/\operatorname{Var}(x)}$ estimates the signal's mean frequency (mobility $\approx 2\pi f_{\mathrm{RMS}}$) from the variances of the signal and its time derivative x'. Because it is obtained from these time-domain variances rather than from the spectral moments, it is not an exact function of f_c and σ_w, and is therefore retained as an independent feature. All retained features (without the specific sub-bands) have previously been applied to signals for bearing condition monitoring and LCM [1, 27]; the Hjorth parameters originate from EEG analysis [28].

Name	**Formula**	**Description**
RMS	$\sqrt{N^{-1}\sum_n x_n^2}$	Overall signal energy in the band; grows with the rate and intensity of contact activity.
Skewness	$N^{-1}\sum_n[(x_n-\bar{x})/\sigma]^3$	Asymmetry of the amplitude distribution; departs from zero when excursions are predominantly one-sided.
Kurtosis	$N^{-1}\sum_n[(x_n-\bar{x})/\sigma]^4$	Impulsiveness; large for sparse, sharp transients and near 3 for Gaussian background noise.
Crest factor	$x_{\mathrm{peak}}/x_{\mathrm{RMS}}$	Prominence of isolated peaks above the background level; elevated by sharp, intermittent events.
Shape factor	$x_{\mathrm{RMS}}\,/\,(N^{-1}\sum_n \lvert x_n\rvert)$	Waveform peakiness independent of amplitude; larger for spiky signals than for smooth, sinusoidal ones.
Margin factor	$x_{\mathrm{peak}}\,/\,(N^{-1}\sum_n \sqrt{\lvert x_n\rvert})^2$	Sensitivity to extreme excursions, weighting large peaks more strongly than the crest factor.
Hjorth mobility	$[\operatorname{Var}(x')/\operatorname{Var}(x)]^{1/2}$	Mean (centroid) frequency of the signal; rises as energy shifts to higher frequencies within the band.
Hjorth complexity	$M(x')/M(x)$	Spectral spread; increases as the signal departs from a single dominant frequency toward broadband content.

Table 2: Time-domain features. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation; $x_{\mathrm{peak}} = (\max x - \min x)/2$ half the peak-to-peak range.

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the sweep. The spectral skewness and kurtosis are standardised by $\sigma_w^p\,\Sigma S$ and are therefore dimensionless and invariant to overall signal amplitude, so that spectral-shape information is separated from the amplitude information carried by RMS.

Name	Formula	Description
Dominant frequency	$f_{\arg\max_k S_k^2}$	Location of the strongest spectral peak; tracks the most energetic resonance or excitation in the band.
Centre frequency	$f_c = \sum_k f_k S_k \,/\, \Sigma S$	Energy-weighted mean frequency; shifts as the balance between low- and high-frequency content changes.
Spectral bandwidth	$\sigma_w = \big(\sum_k (f_k - f_c)^2 S_k \,/\, \Sigma S\big)^{1/2}$	Concentration of energy about the centroid; small for tonal signals, large for broadband activity.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k \,/\, (\sigma_w^3\, \Sigma S)$	Direction of spectral asymmetry; indicates whether energy leans below or above the centroid.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k \,/\, (\sigma_w^4\, \Sigma S)$	Tailedness of the spectrum; large when energy is spread out.
Spectral flatness	$\exp(K'^{-1} \sum_{k:S_k>0} \ln S_k) \,/\, \bar{S}$	Tonal-versus-noise character; near 1 for broadband noise, near 0 for a pure tone.

Table 3: Frequency-domain features (companion to Table 2). Notation: $S_k = |X_k|$ one-sided FFT magnitude at bin k; f_k corresponding frequency; $\bar{S}$ arithmetic mean of $\{S_k\}$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1} \sum_k f_k S_k$; $\sigma_w = \big((\Sigma S)^{-1} \sum_k (f_k - f_c)^2 S_k\big)^{1/2}$.

4.5 Candidate feature sets

Extracting the 14-feature set independently for the broadband signal and each sub-band yields $14 \times 3 = 42$ candidate AE features (broadband plus two sub-bands) and $14 \times 4 = 56$ candidate US features (broadband plus three sub-bands) per sweep. Features are named by band and statistic (e.g. `AE_500-1000kHz__mobility`); broadband features carry no band prefix. These two candidate sets, and their union (the combined AE+US set, $42 + 56$ features), define the three sensor configurations—AE-only, US-only, and combined—evaluated in Sections 5–6. The reduction of these candidate sets to compact, non-redundant subsets for modelling is described in Section 5.

5 Feature Analysis

This section reduces the 42 AE and 56 US candidate features (Section 4) to compact, non-redundant subsets for modelling, and characterises what the surviving features respond to. The analysis proceeds in two selection stages – a relevance filter against κ followed by a redundancy reduction – after which the operating-condition dependence and low-dimensional structure of the retained sets are examined.

5.1 Two-stage feature selection

Stage 1 – relevance. Each candidate feature is ranked by its absolute Spearman ($|\rho|$, monotonic association) and Pearson ($|r|$, linear association) correlation with κ across all sweeps. Features must satisfy both $|\rho| \geq 0.1$ and $|r| \geq 0.1$ to be retained, removing features carrying essentially no monotonic or linear information about the target.

Stage 2 – redundancy. Among the survivors, multicollinearity is reduced by iterative variance-inflation-factor (VIF) elimination. At each round the VIF of every remaining feature is computed; if any feature exceeds the threshold VIF $= 5$, the *least κ-relevant* feature among those above the threshold (lowest $|\rho|$ with κ) is dropped, and all VIFs are recomputed on the reduced set. The procedure repeats until no remaining feature exceeds the threshold. Recomputing the VIFs each round addresses both pairwise and distributed (many-to-one) collinearity, while the relevance-based tie-break protects the strongest predictor in each collinear group: a highly inter-correlated feature is retained when it is the most κ-relevant member of its group, its less-relevant partners being removed in its place. By construction every retained feature has VIF < 5 on the final subset.

Table 4 summarises the selection funnel.

Table 4: Feature selection funnel: candidate features, survivors of the Stage 1 relevance filter, and the final retained sets after Stage 2 redundancy reduction.

Channel	**Candidates**	**After Stage 1**	**Retained**
AE	42	36	11
US	56	35	12
Combined	–	–	23

5.2 Correlation with κ

Table 5 lists the top ten features for each sensor ranked by combined Spearman–Pearson rank score with κ; the complete rankings of all candidates, including Stage 2 retention, are given in Appendix A (Tables 11 and 12).

AE			US		
Feature	$\lvert\rho\rvert$	$\lvert r\rvert$	Feature	$\lvert\rho\rvert$	$\lvert r\rvert$
500 kHz–1 MHz mobility†	0.848	0.662	10–20 kHz RMS	0.691	0.609
500 kHz–1 MHz centre freq.	0.799	0.687	RMS	0.614	0.547
Shape factor†	0.703	0.625	0–10 kHz RMS	0.610	0.542
500 kHz–1 MHz dominant freq.	0.676	0.582	10–20 kHz mobility†	0.583	0.536
500 kHz–1 MHz complexity	0.682	0.529	20–100 kHz mobility	0.420	0.466
20–500 kHz shape factor	0.631	0.554	20–100 kHz RMS	0.507	0.344
500 kHz–1 MHz RMS†	0.617	0.528	Complexity†	0.542	0.297
20–500 kHz centre freq.	0.563	0.543	20–100 kHz complexity†	0.416	0.396
Centre freq.	0.577	0.524	0–10 kHz kurtosis†	0.465	0.331
20–500 kHz mobility	0.532	0.550	Kurtosis	0.452	0.300

Table 5: Top 10 features by combined Spearman–Pearson rank score with κ, for the AE and the US signals. Features retained after Stage 2 redundancy reduction are marked with †.

The top-ranked AE feature is 500 kHz–1 MHz mobility ($|\rho| = 0.848$), and the top-ranked US feature is 10–20 kHz RMS ($|\rho| = 0.691$).

Two patterns distinguish the channels. For AE, relevance concentrates in the 500 kHz–1 MHz sub-band, which supplies five of the ten highest-ranked features (including the top-ranked mobility) and is led by spectral-shape and frequency-distribution descriptors—mobility, centre and dominant frequency, complexity—rather than by raw amplitude: where the high-frequency energy sits and how peaked its spectrum is carry more monotonic information about κ than its magnitude alone. For US, relevance instead concentrates in the low-frequency sub-bands ($\leq$ 20 kHz) to where it has been heterodyned and is dominated by amplitude (RMS) features, with the broadband and 0–20 kHz RMS terms heading the ranking. The channels also differ in strength: the best AE feature exceeds the best US feature by a wide margin in $|\rho|$ (0.848 vs. 0.691), consistent with the AE sensor's more direct mechanical coupling to asperity-contact events. Cross-referencing the retained features against operating conditions (Section 5.4) shows that the most κ-relevant AE features are also those most strongly correlated with rotational speed, with one informative exception: the 20–500 kHz skewness feature is almost uncorrelated with speed yet carries the strongest temperature association of the retained set, anticipating the within-step temperature channel examined below.

Broadband AE features rank lower than their sub-band counterparts, indicating that the frequency-specific content of the high-frequency AE signal carries more monotonic information about κ than the full-band aggregate [7, 22].

5.3 Redundancy structure

Figure 7 shows the variance inflation factors of the Stage 1 survivors, computed on the full survivor set, on a logarithmic axis. The values span several orders of magnitude. A feature plotted far above the threshold may still be retained: the iterative procedure (Section 5.1) removes its less κ-relevant collinear partners first, so its *recomputed* VIF on the reduced set falls below the threshold even though its full-set value is large.

The redundancy concentrates in the amplitude-ratio family. RMS, crest factor, margin factor, and shape factor are algebraically interrelated normalisations of the same waveform amplitude, and across sub-bands they form a tightly collinear cluster whose members carry the largest variance inflation factors—several orders of magnitude above the threshold. The iterative elimination therefore prunes this cluster most heavily, removing the duplicated band-copies and lower-relevance amplitude ratios until the surviving amplitude features are mutually non-collinear. A secondary, weaker cluster links the spectral-location descriptors (centre frequency, dominant frequency, bandwidth) within a band. By contrast, the distributional-shape features—skewness and spectral kurtosis—are close to mutually orthogonal and enter the retained set almost untouched. Because the candidate set is canonical (Section 4), these collinearities reflect genuine shared structure—primarily the common amplitude scaling imposed by the operating point—rather than definitional identities among overlapping feature definitions.

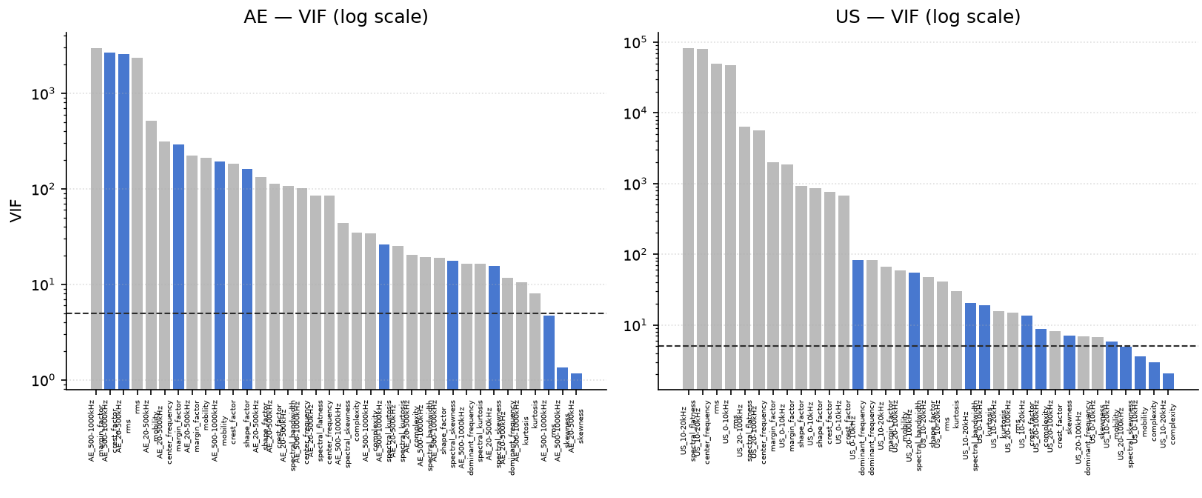

Figure 7: Variance inflation factors of the Stage 1 survivors, computed on the full survivor set (logarithmic axis). Retained features in blue, removed features in grey; the dashed line marks the VIF = 5 threshold of the Stage 2 iterative elimination. Values are capped at 10^6 for display.

5.4 Operating-condition dependence

The raw correlation between signal features and κ is partially driven by the shared dependence on rotational speed and temperature. Higher RPM reduces the required viscosity ν_1

(Eq. (2)) and therefore increases κ, while simultaneously increasing signal energy through more frequent contact events per second. Higher temperature reduces the actual viscosity ν and therefore reduces κ, while also affecting material damping and structural dynamics.

As a consequence, amplitude features such as RMS are partially acting as indirect proxies for rotational speed rather than as direct indicators of lubrication film adequacy [7, 17, 21]. This confounding is inherent to any single-operating-point sensor system without an explicit operating-condition normalisation step. Because κ is fully determined by the operating point (Section 2.1), a model given measured RPM and temperature directly would predict κ near-perfectly by construction; the contribution of the acoustic features is to recover this information without tachometer or thermometer access.

Marginally, the retained features correlate with κ chiefly through this operating-point dependence. Rotational speed is the dominant axis—it has a Spearman correlation of $\rho = +0.75$ with κ, against $\rho = -0.61$ for temperature—and a LightGBM regressor on the retained AE features recovers the operating point itself almost completely (RPM at $R^2 = 0.945$, temperature at $R^2 = 0.663$). Passing these predicted conditions through Eqs. (1)–(2) yields a two-stage κ estimate ($R^2 = 0.826$) that nearly equals the direct features-to-κ model ($R^2 = 0.834$), so virtually all of the marginal skill is attributable to acoustic operating-point proxying rather than to direct film sensing. This marginal view, however, is dominated by the speed sweep and conflates the two κ channels. We therefore condition on each operating variable in turn, using the measured operating point: the within-RPM-step correlation with temperature (sweeps grouped into staircase steps of fixed speed) isolates the temperature response, while the within-temperature-block correlation with speed (sweeps grouped into 2 °C blocks of measured temperature, within which the temperature standard deviation stays below 1 °C) isolates the speed response. Figure 8 contrasts the marginal and conditional structure for every retained feature of both sensors. The speed axis is essentially unchanged by conditioning—the features genuinely track rotational speed—whereas the temperature axis spreads outward once speed is removed, uncovering a lubrication-state sensitivity that the marginal view masks; the strongest AE feature, 20–500 kHz mobility, reaches a within-step correlation of $\rho = -0.56$ with temperature. The features are thus acoustic operating-point soft sensors that, beyond the proxy share, retain a genuine residual sensitivity to the lubrication state.

The two sensors occupy distinct regions of this conditional plane. AE features cluster on the speed axis, whereas the passive-US channel resolves the temperature dependence more strongly and with mixed sign across its sub-bands (Figure 8, right). AE and US therefore capture the lubrication state through partly distinct channels rather than redundantly—the complementarity their fusion later exploits.

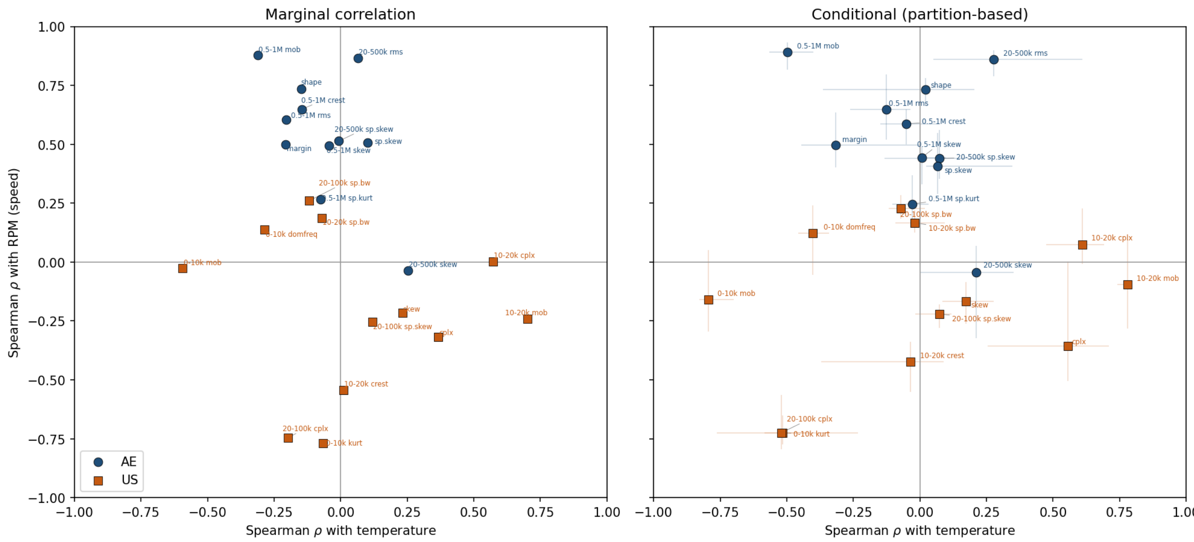

Figure 8: Marginal versus conditional rank-correlation structure of the retained AE (circles) and US (squares) features. Each point is one feature; the axes are the Spearman correlation with measured temperature (x) and rotational speed (y). *Left:* marginal correlations over all sweeps, dominated by the speed sweep. *Right:* partition-based conditional correlations—within-RPM-step for the temperature axis and within-temperature-block for the speed axis—with whiskers showing the inter-quartile range across partitions. Conditioning leaves the speed axis essentially unchanged but spreads the features along the temperature axis, revealing lubrication-state sensitivity masked in the marginal view. AE features track speed; US features lean toward temperature.

5.5 Low-dimensional structure and regime separability

Figure 9 shows the first two principal components of the retained AE and US feature sets coloured by κ value. In the AE projection the κ value varies smoothly across the plane rather than forming disjoint clusters: high-κ sweeps (well-separated films) concentrate in the upper-left lobe while low-κ sweeps fan out along a tail extending to large positive PC1, so the first two components—together capturing roughly two-thirds of the AE feature variance—already arrange the data along a continuous lubrication gradient. The US projection shows the same ordering more weakly: a single dense core carries a shallower κ gradient and a detached low-κ cluster sits apart from it, consistent with the lower predictive power of the US channel (Section 6). The continuity of the AE gradient, rather than sharp regime boundaries, mirrors the continuous nature of κ and indicates that the retained features encode a graded film-thickness signal rather than a small number of discrete operating states.

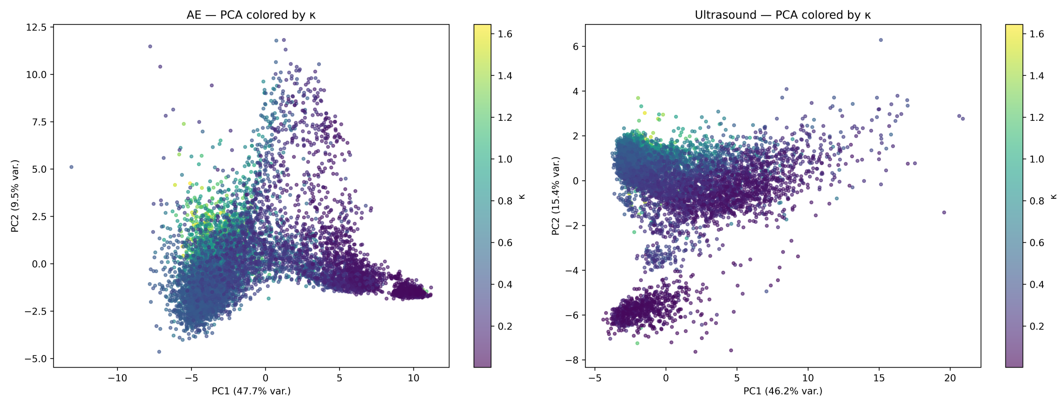

Figure 9: PCA projection of the retained AE and US feature sets, coloured by lubrication κ value.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design

6.1.1 Data sets and splits

The data experiments utilise three different features sets: the AE, the US, and the combined AE and US feature sets. Each of the later described models are fitted to each of the three datasets. Each dataset is divided into a training set (80%) and a hold-out test set (20%) by a *visit-grouped* split: sweeps are partitioned into groups corresponding to contiguous 60-second RPM holds (both sensor rows of a sweep kept together), and whole groups—rather than individual sweeps—are assigned to the training or hold-out set with `GroupShuffleSplit`. This prevents the near-duplicate windows of a single slowly-varying hold from straddling the train/test boundary, as a sweep-level random split would allow. The hold-out set is withheld entirely from all model development steps and used only for final evaluation. The training set is further divided during nested cross-validation described in Section 6.2.

6.1.2 Model families

Three regression model families are evaluated for three sensor configurations, as summarised in Table 6.

6.2 Repeated nested cross-validation

To obtain stable performance estimates that properly account for the variance introduced by the train/validation split, the model evaluation uses a repeated nested k-fold cross-validation scheme. The outer loop is repeated $R = 5$ times with independently re-seeded, group-respecting 5-fold splits (the same hold-group structure as the hold-out split, so

Model	Rationale	Hyperparameters & tuning
Elastic Net	$\ell_1+\ell_2$ regularised linear model; ℓ_1 promotes sparsity, ℓ_2 stabilises correlated features.	α (9 values, 10^{-4}–10^2) $\times$ ℓ_1 ratio (6 values, 0.1–1.0); `ElasticNetCV`, inner 5-fold.
Polynomial (deg. 2)	Augments linear model with all pairwise interaction terms; remains linear in parameters.	Ridge penalty λ (15 log-spaced values, 10^{-3}–10^4); `RidgeCV` (leave-one-out) within each outer fold.
LightGBM	Gradient-boosted trees; captures non-linear feature–κ mappings and high-order interactions.	Learning rate, num. leaves, max. depth, min. child samples, ℓ_2 reg.; Bayesian optimisation (Optuna TPE, 20 trials per outer fold).

Table 6: Regression model families: rationale and hyperparameter tuning strategy.

near-duplicate windows never straddle a fold boundary), yielding $R \times k = 25$ outer RMSE scores per model–sensor combination. See Figure 10 for data partitions.

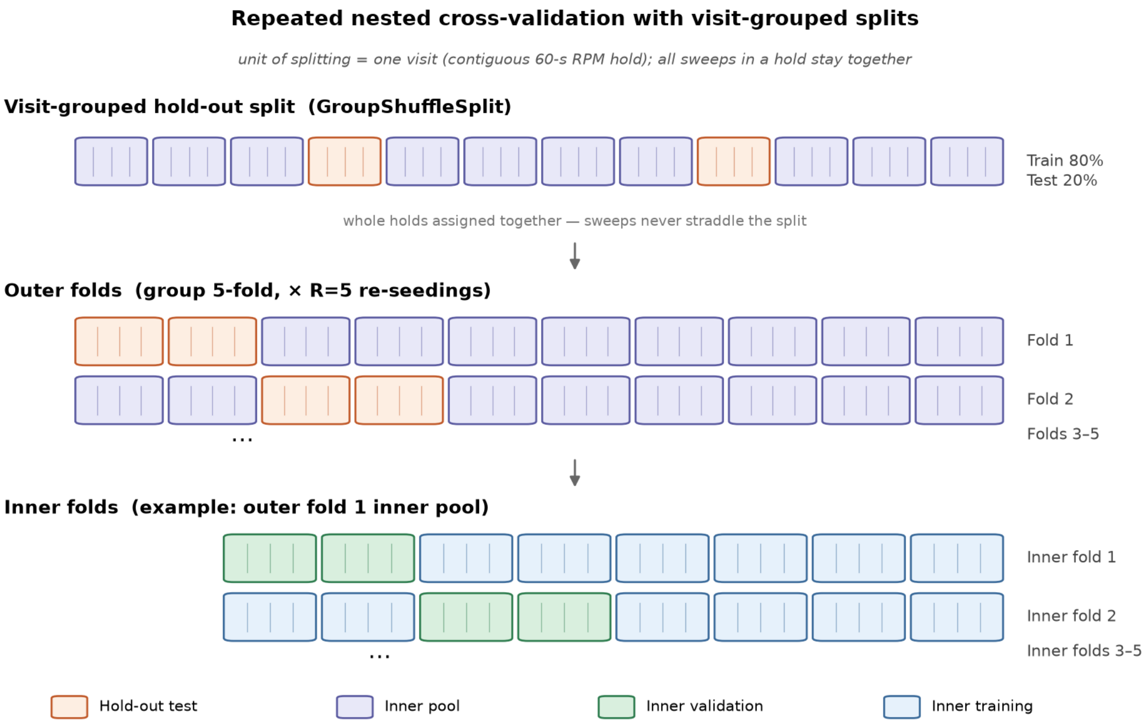

Figure 10: Data partitioning for the repeated nested cross-validation: an 80/20 group-level hold-out split (groups = 60-second RPM holds), with the training pool further divided into $R = 5$ independently seeded, group-respecting 5-fold outer splits and inner hyperparameter-selection folds.

For LightGBM, each outer fold contains its own inner Optuna optimisation loop, ensuring that the reported outer scores are not contaminated by hyperparameter selection on the outer fold data. For the linear models, hyperparameter selection runs within each outer training partition: `ElasticNetCV` performs an inner 5-fold search over the α/ℓ_1-ratio grid, and `RidgeCV` a leave-one-out generalised cross-validation over the λ grid; the selected hyperparameters are then refit on the full outer-training fold before the outer validation fold is scored. These inner resampling schemes are not group-aware, so near-duplicate sweeps from the same hold may render their hyperparameter scores mildly optimistic; this affects only the selected penalties, as final performance is scored on the visit-grouped outer folds.

The 25 outer scores form the empirical sampling distribution of RMSE for each model and are used directly in the significance tests described below.

6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels, with the batteries summarised in Table 7. *Within-sensor* comparisons (architecture differences, e.g. Elastic Net vs. Polynomial on AE) ask which model generalises better and are assessed on the 25 repeated nested-CV fold RMSEs. *Cross-sensor* comparisons (the best model per sensor configuration, selected by lowest mean CV RMSE) are the study's focal claims and are assessed both on the CV fold RMSEs and on per-sample residuals from the common hold-out set. The Nadeau–Bengio corrected t-test [29] gauges robustness across resampling partitions—correcting the paired t-test for the positive correlation induced by training-set overlap across folds—while the hold-out tests gauge the difference on held-out data and the bootstrap supplies an effect size. All p-values are Holm–Bonferroni-corrected within each family at $\alpha = 0.05$; where CV and hold-out evidence diverge, both are reported.

Test	Unit of analysis	Within-sensor	Cross-sensor
Nadeau–Bengio corrected t-test	nested-CV fold RMSEs	✓	✓
Wilcoxon signed-rank	hold-out per-sample residuals	—	✓
Diebold–Mariano	hold-out per-sample residuals	—	✓
Bootstrap ΔRMSE 95% CI	hold-out residuals (10 000 resamples)	—	✓

Table 7: Significance-testing protocol. Within-sensor comparisons (model architectures) use the corrected repeated-CV t-test on fold scores; cross-sensor comparisons (sensor and fusion contrasts) additionally use hold-out residual tests and a bootstrap effect-size interval. All p-values are Holm–Bonferroni-corrected within each family; divergences between CV and hold-out evidence are reported in full.

6.4 Results

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0876 ± 0.0203	**0.908**	0.056	0.081
LightGBM	AE	0.1138 ± 0.0177	0.834	0.082	0.109
LightGBM	US	0.1648 ± 0.0372	0.653	0.105	0.158
Polynomial	Combined	0.1848 ± 0.0416	0.677	0.116	0.153
Elastic Net	Combined	0.1913 ± 0.0434	0.593	0.134	0.171
Polynomial	AE	0.2027 ± 0.0381	0.487	0.155	0.192
Polynomial	US	0.2236 ± 0.0552	0.333	0.163	0.219
Elastic Net	AE	0.2268 ± 0.0455	0.491	0.152	0.192
Elastic Net	US	0.2279 ± 0.0525	0.463	0.149	0.197

Table 8: Regression model performance. CV RMSE: mean ± std of outer RMSE across 25 repeated nested CV folds (training set $n = 7\,178$). HO: hold-out test set ($n = 1\,326$). Best hold-out R^2 in bold.

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations, with AE-only reaching $R^2 = 0.834$, RMSE $= 0.109$ κ-units. On a target range of [0.04, 1.64] with mean 0.51, this corresponds to a relative RMSE of approximately 21% of the mean target value. Linear and polynomial models on AE features achieve $R^2 = 0.487$–0.491, confirming a non-linear feature–κ mapping that gradient boosting exploits more effectively. The within-sensor corrected repeated nested-CV t-test (Table 7) confirms this gap is significant: LightGBM outperforms both linear families on AE (vs. Elastic Net $p = 9.8 \times 10^{-4}$, vs. Polynomial $p = 6.6 \times 10^{-4}$; Holm-corrected), whereas the two linear families are statistically indistinguishable.

Sensor comparison. AE features substantially outperform US features across all model families. The LightGBM AE–US gap ($\Delta R^2 = 0.18$) is large and unambiguous on the hold-out set, although the corrected repeated nested-CV t-test does not reach significance; we report both (Table 9). This reflects the richer frequency content and more direct physical sensitivity of the AE sensor to asperity contact events.

Sensor fusion. LightGBM on the combined AE+US feature set achieves $R^2 = 0.908$ ($\Delta R^2 = 0.074$ over the AE-only result of 0.834). The gain is significant under both the corrected repeated nested-CV t-test and all hold-out tests (Table 9). For the linear models, the combined feature set performs better than AE alone (Elastic Net: R^2 0.593 vs. 0.491; Polynomial: 0.677 vs. 0.487). Although the absolute R^2 gain is modest, this is a relative RMSE reduction of about 26% over AE alone, consistent across all four tests, indicating that the US channel contributes genuine, statistically robust non-redundant information.

Comparison	Nadeau–Bengio p	Wilcoxon p	Diebold–Mariano p	ΔRMSE [95% CI]
AE vs. US	0.121	1.7×10^{-7}	7.2×10^{-12}	0.0487 [0.0368, 0.0612]
AE vs. Combined	5.1×10^{-6}	9.6×10^{-39}	6.5×10^{-18}	0.0281 [0.0219, 0.0343]
US vs. Combined	0.002	3.0×10^{-75}	4.8×10^{-26}	0.0768 [0.0666, 0.0875]

Table 9: Cross-sensor inference for the best model per configuration (LightGBM). The Nadeau–Bengio corrected t-test is computed on the repeated nested-CV fold RMSEs; the Wilcoxon, Diebold–Mariano, and bootstrap ΔRMSE interval on the common hold-out residuals. All p-values are Holm–Bonferroni-corrected within each family. The AE–US row illustrates the divergence noted in the text: the gap is decisive on the hold-out tests yet not significant under the corrected repeated-CV t-test ($p = 0.121$), hence treated as inconclusive at the fold level.

Model consistency. The close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting.

Figure 11 shows predicted vs. actual κ on the hold-out set for all model families and sensor configurations, coloured by RPM.

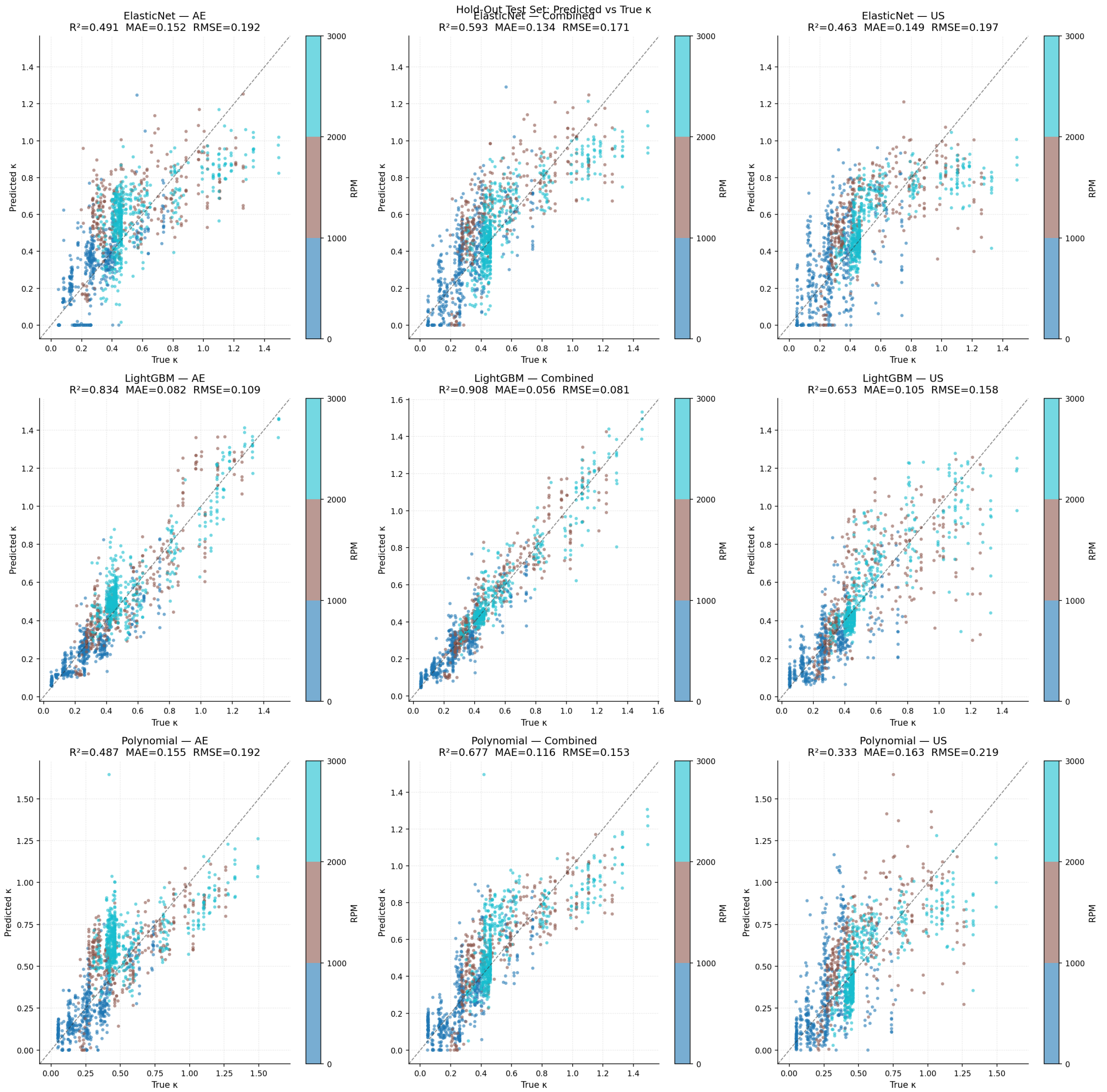

Figure 11: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations, colored according to RPM. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

Across the grid, predictions track the diagonal most closely for LightGBM—tightest in the AE and combined configurations—whereas the Elastic Net and polynomial families systematically under-predict at high κ, their points bowing below the diagonal. For every model the residual spread widens in the mixed-lubrication regime ($0.5 \leq \kappa < 1.0$) and toward the under-represented full-film range ($\kappa > 1$); the RPM colouring reflects the operating-point dependence of κ (Section 5.4).

SHAP-based feature importance. SHAP (SHapley Additive exPlanations) values computed via TreeExplainer identify which features drive the κ predictions for each sensor (Figure 12). For the AE LightGBM model the dominant contributor by a large margin is *500 kHz–1 MHz mobility* (mean $|\text{SHAP}| = 0.205$), followed by *20–500 kHz RMS*

(0.050) and *500 kHz–1 MHz crest factor* (0.036); this 500 kHz–1 MHz mobility feature ranks *first* for both LightGBM and the polynomial model and third for Elastic Net, the most robust single AE predictor across model families. The US model is led instead by *10–20 kHz mobility* (0.128), then *20–100 kHz complexity* (0.101) and *0–10 kHz mobility* (0.069). The two channels thus rely on distinct features whose operating-condition leanings differ—the dominant AE feature is speed-tracking, whereas the dominant US feature is the temperature-leaning 10–20 kHz mobility (Figure 8)—so SHAP independently corroborates their separate-channel structure. (The US band labels refer to the probe's heterodyned audio baseband, not raw acoustic frequencies, so the contrast is one of operating-condition channel rather than frequency.) Cross-model agreement for both sensors is summarised in Table 10.

Feature	**ElasticNet**	**Polynomial**	**LightGBM**	**In top 5**
AE				
500 kHz–1 MHz mobility	3	1	1	3
Broadband shape factor	1	4	4	3
20–500 kHz RMS	2	–	2	2
500 kHz–1 MHz crest factor	4	–	3	2
Broadband margin factor	5	2	–	2
US				
10–20 kHz mobility	2	2	1	3
0–10 kHz mobility	1	1	3	3
20–100 kHz complexity	3	4	2	3
0–10 kHz kurtosis	–	3	4	2
Broadband complexity	5	–	5	2

Table 10: Cross-model SHAP agreement, grouped by sensor: base features that rank among the 5 most important (by mean |SHAP|) for at least two of the three model families. Polynomial ranks are positions in the full expanded-term importance list (interaction and squared terms included); only base features are tabulated.

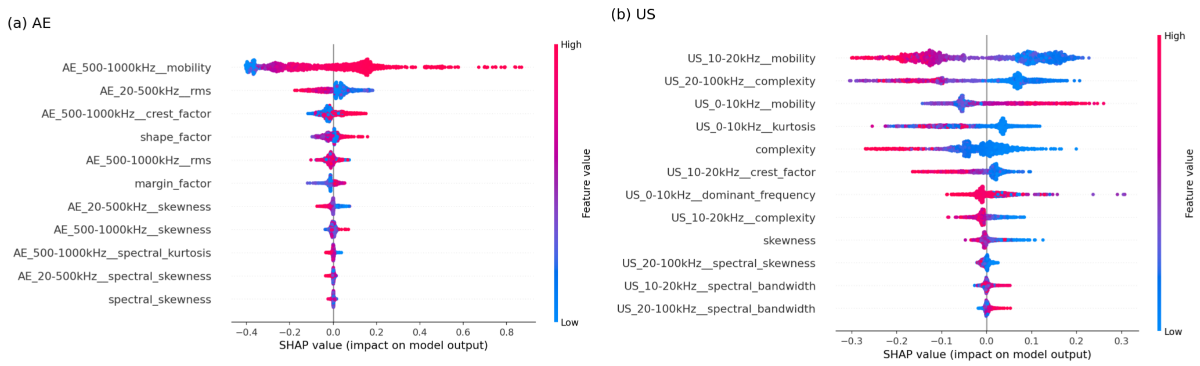

Figure 12: SHAP beeswarm of per-sample feature contributions for the LightGBM model on (a) AE and (b) US features; features are ordered by mean |SHAP value| and coloured by feature value. The dominant AE feature (500 kHz–1 MHz mobility) tracks speed, whereas the dominant US feature (10–20 kHz mobility, on the probe's heterodyned baseband) is temperature-leaning—mirroring the operating-condition channel separation of Figure 8.

7 Discussion

7.1 AE vs. US

The consistently superior performance of AE over the US channel across all model might reflect a fundamental hardware difference. The AE sensor captures wideband content extending into the MHz range (analysed here up to 1 MHz), dominated by verified stress-wave emission up to several hundred kHz. The US channel down-mixes its ≈38–40 kHz sensing band to an audio-frequency baseband (Section 2.2). Although the probe acquires content across its 0–100 kHz band (Section 4.2), the lubrication-relevant information is concentrated in the nominal 0–20 kHz baseband—a mixture of down-mixed contact activity, low-frequency machinery noise, VFD harmonics, and structural vibration, within which the 10–20 kHz half carries the strongest κ correlations (Section 5.2)—while the 20–100 kHz above-baseband content contributes comparatively little. The high-frequency transient signature of individual microscale contact events, which the AE channel is thought to capture directly, is largely absent from this down-mixed envelope. Replacing the heterodyned probe with a wideband passive acoustic or ultrasound sensor would be expected to substantially improve the US-channel models.

7.2 Fusion gain

Fusing the US channel into the AE model reduces hold-out RMSE from 0.109 to 0.081 (ΔRMSE $= 0.0281$; $\Delta R^2 = 0.074$)—a relative reduction of about 26%, statistically robust under repeated nested cross-validation and confirmed by all three hold-out tests. Although the absolute R^2 gain is small, against an AE baseline of $R^2 = 0.834$ it is a substantial cut in the residual error, confirming that the US channel adds genuinely non-redundant information. That contribution is nonetheless bounded by the relative weakness of the US channel, which retains only 12 features after selection (vs. 11 for AE) and whose top correlations ($|\rho| \leq 0.691$) are weaker than AE's ($|\rho| = 0.848$).

This result is an informative contribution to the LCM literature, where AE–US complementarity is *argued* but rarely *tested*. The conditional correlation structure (Figure 8) illustrates this complementarity at the feature level: the two channels lean toward different operating-condition axes—AE toward speed, US toward temperature—consistent with their carrying partly non-redundant information, even if the magnitude of that independence is modest. Whether a wideband passive ultrasound sensor (rather than a heterodyned probe) would provide more independent information – and thus a more substantial fusion gain – remains an open question for future work.

7.3 Interpreting the amplitude-confounding result

The strong raw Spearman correlations of amplitude features (RMS and related statistics) and frequency-tracking features (Hjorth mobility, centre frequency) with κ should be interpreted with caution. As discussed in Section 5.4, rotational speed is a driver of both κ (through ν_1 in Eq. (2)) and broadband signal energy (through the rate of and energy in contact events per revolution as described in [7]). A model trained on amplitude features therefore exploits speed as an indirect proxy for κ rather than sensing purely lubrication-specific signal changes directly.

This confounding does not invalidate the regression results: a model that uses speed-mediated amplitude changes to predict κ is still predicting κ, and its predictions may be operationally useful. However, it does indicate that the model's generalisation to operating conditions outside its training distribution (different speed ranges, different bearing loads) may be limited. The two-stage analysis of Section 5.4 makes this concrete: predicted operating points alone, fed through the ISO 281 mapping, recover $R^2 = 0.826$ of the direct model's $R^2 = 0.834$—the model is an operating-point soft sensor, and its lubrication sensitivity beyond the operating point rests on the within-step evidence of Section 5.4. Future feature engineering should prioritise amplitude-invariant spectral-shape features (spectral skewness and kurtosis, spectral bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information relative to operating condition proxies.

7.4 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM, when κ is assumed to be representative of the lubrication conditions in the bearing. A hold-out $R^2 = 0.834$ and RMSE = 0.109 κ-units for a single AE sensor represents strong performance across a demanding continuous speed-and-temperature ramp spanning boundary through full-film lubrication conditions. Under matched random (sweep-level) splits this performance is on par with the passive κ regression reported by Jakobsen et al. [1, 16]. The figures reported here are not directly comparable to most prior work, however, because they are obtained under a stricter *visit-grouped* evaluation protocol (Section 6.1) that removes the near-duplicate leakage admitted by a sweep-level random split; to our knowledge they are the first κ-regression results reported under such a leakage-controlled protocol. The contribution is thus methodological—a more conservative, deployment-representative performance estimate—rather than a raw accuracy gain over earlier studies.

The repeated nested cross-validation design provides appropriately conservative performance estimates: the 25 outer RMSE scores capture fold-to-fold variance without the optimistic bias introduced by simple k-fold evaluation with shared hyperparameter search. The corrected significance tests confirm that performance differences between model families are assessed under assumptions that properly account for data re-use.

7.5 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown.

2. **Heterodyned US channel.** The passive US probe down-mixes a narrow ≈38–40 kHz sensing band to an audio-frequency baseband rather than recording wideband ultrasound directly, and its lubrication-relevant information is concentrated below ≈20 kHz (Section 2.2). It should not be equated with a wideband passive ultrasound sensor, and the US-channel results should be interpreted in light of this hardware constraint.

3. **Skewed κ coverage.** The sampled κ distribution is skewed toward the boundary and mixed-lubrication regime, with full-film states ($\kappa > 1$) under-represented and the highest κ values reached only rarely (Figure 2). Extrapolation into the undersampled full-film regime should therefore be treated with caution.

4. **Model-derived κ reference.** The viscosity ratio is not measured directly but computed for each sweep from the measured operating point—rotational speed and housing temperature—via the ISO 281 relations and the ASTM D341 Walther viscosity–temperature equation (Section 2.1). The regression target is therefore a model-derived reference that inherits the assumptions of those relations, notably the use of housing temperature as a proxy for the elastohydrodynamic contact temperature; this also underlies the operating-point soft-sensor interpretation of Section 7.3.

8 Conclusion

This paper has presented a controlled study of simultaneous acoustic emission and passive ultrasound sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **AE sub-band features are predictors of κ.** LightGBM on 11 retained AE features achieves hold-out $R^2 = 0.834$, RMSE $= 0.109$ κ-units. The 500 kHz–1 MHz mobility feature is the dominant SHAP contributor and the most robust single predictor across model families—ranked first for the LightGBM and polynomial models (third for Elastic Net)—identifying the upper AE sub-band (500 kHz–1 MHz) as the most informative frequency range for lubrication-regime prediction.

2. **Sensor fusion yields consistently significant gain.** The LightGBM combined model achieves $R^2 = 0.908$ ($\Delta R^2 = 0.074$ over AE alone); within-CV corrected t-test $p = 5.1 \times 10^{-6}$; hold-out tests: Wilcoxon $p = 9.6 \times 10^{-39}$, Diebold–Mariano $p = 6.5 \times 10^{-18}$, bootstrap 95% CI [0.0219, 0.0343] excluding zero. The heterodyned US channel thus supplies a modest amount of non-redundant information, though its 12 retained features are individually weaker than the AE features.

3. **AE outperforms the US channel.** The AE–US gap ($\Delta R^2 = 0.18$ for LightGBM) is significant on the hold-out tests (Wilcoxon $p = 1.7 \times 10^{-7}$), although the corrected repeated-CV t-test does not reach significance ($p = 0.121$); we therefore report both. The gap might reflect the bandwidth limitation of the heterodyned US probe.

4. **AE and US features are informative about lubrication conditions through different paths.** The AE features are mainly informative on rotational speed, while US features are more informative on temperature. This decomposition and complementarity aides understanding and development of data-driven multi-modal methods for LCM.

5. **Rigorous evaluation via repeated nested cross-validation.** The 25 outer RMSE scores per model provide stable, conservative performance estimates; corrected significance tests confirm that LightGBM significantly outperforms both linear families within each sensor configuration (Holm–Bonferroni corrected), while the two linear families are statistically indistinguishable.

6. **The κ regression framework is workable for passive non-invasive LCM, with a quantified caveat.** Most of the headline performance is recoverable from acoustically inferred operating points alone ($R^2 = 0.826$ via the two-stage decomposition), so the framework is best understood as an acoustic operating-point soft sensor combined with the ISO 281 mapping; the within-speed-step analysis nevertheless

confirms a genuine, temperature-mediated film channel beyond the operating-point proxy.

Future work should address the identified limitations by: validating the model-derived κ reference against an independent film-thickness proxy; acquiring wideband passive ultrasound to replace the heterodyned probe; validating across multiple independent sessions, bearing types, and lubricants; running bearings for longer to degrade the lubrication conditions through other means than nominal operating conditions for a more realistic data-set; and developing physically motivated sub-band ratio features that are explicitly designed to be robust to operating-condition variation.

A Full Feature Correlation Rankings

Tables 11 and 12 list the complete Spearman and Pearson correlations with κ for all AE and US features that passed the initial data quality filter, sorted by combined rank. Features retained after the Stage 2 redundancy reduction are indicated with a checkmark.

Table 11: Full AE feature correlation ranking with κ. $|\rho|$: absolute Spearman correlation; $|r|$: absolute Pearson correlation. Retained: selected after Stage 2 redundancy reduction.

Rank	Sub-band	Feature	$\lvert\rho\rvert$	$\lvert r\rvert$	Ret.
1	500–1000kHz	mobility	0.848	0.662	✓
2	500–1000kHz	center frequency	0.799	0.687	–
3	Broadband	shape factor	0.703	0.625	✓
4	500–1000kHz	dominant frequency	0.676	0.582	–
5	500–1000kHz	complexity	0.682	0.529	–
6	20—-500 kHz	shape factor	0.631	0.554	–
7	500–1000kHz	rms	0.617	0.528	✓
8	20—-500 kHz	center frequency	0.563	0.543	–
9	Broadband	center frequency	0.577	0.524	–
10	20—-500 kHz	mobility	0.532	0.550	–
11	500–1000kHz	crest factor	0.553	0.523	✓
12	Broadband	margin factor	0.589	0.485	✓
13	Broadband	mobility	0.533	0.535	–
14	500–1000kHz	margin factor	0.509	0.512	–
15	Broadband	complexity	0.566	0.454	–
16	20—-500 kHz	complexity	0.523	0.473	–
17	20—-500 kHz	rms	0.603	0.339	✓
18	Broadband	rms	0.601	0.320	–

(continued)

Rank	Sub-band	Feature	$\lvert\rho\rvert$	$\lvert r\rvert$	Ret.
19	20—-500 kHz	dominant frequency	0.381	0.521	–
20	500–1000kHz	kurtosis	0.469	0.404	–
21	20—-500 kHz	margin factor	0.464	0.343	–
22	Broadband	crest factor	0.504	0.288	–
23	Broadband	kurtosis	0.491	0.174	–
24	20—-500 kHz	spectral kurtosis	0.390	0.267	–
25	20—-500 kHz	spectral skewness	0.372	0.220	✓
26	500–1000kHz	shape factor	0.296	0.289	–
27	500–1000kHz	skewness	0.364	0.198	✓
28	500–1000kHz	spectral kurtosis	0.298	0.258	✓
29	20—-500 kHz	crest factor	0.371	0.157	–
30	20—-500 kHz	skewness	0.209	0.261	✓
31	Broadband	spectral skewness	0.284	0.174	✓
32	20—-500 kHz	spectral flatness	0.246	0.165	–
33	20—-500 kHz	spectral bandwidth	0.241	0.164	–
34	500–1000kHz	spectral skewness	0.198	0.157	–
35	Broadband	spectral kurtosis	0.229	0.108	–
36	500–1000kHz	spectral bandwidth	0.135	0.105	–

Table 12: Full US feature correlation ranking with κ. $|\rho|$: absolute Spearman correlation; $|r|$: absolute Pearson correlation. Retained: selected after Stage 2 redundancy reduction.

Rank	Sub-band	Feature	$\lvert\rho\rvert$	$\lvert r\rvert$	Ret.
1	10–20kHz	rms	0.691	0.609	–
2	Broadband	rms	0.614	0.547	–
3	0–10kHz	rms	0.610	0.542	–
4	10–20kHz	mobility	0.583	0.536	✓
5	20–100kHz	mobility	0.420	0.466	–
6	20–100kHz	rms	0.507	0.344	–
7	Broadband	complexity	0.542	0.297	✓
8	20–100kHz	complexity	0.416	0.396	✓
9	0–10kHz	kurtosis	0.465	0.331	✓
10	Broadband	kurtosis	0.452	0.300	–
11	10–20kHz	crest factor	0.350	0.394	✓
12	0–10kHz	mobility	0.380	0.348	✓
13	10–20kHz	complexity	0.342	0.306	✓

(continued)

Rank	Sub-band	Feature	$\|\rho\|$	$\|r\|$	Ret.
14	10–20kHz	margin factor	0.280	0.358	–
15	20–100kHz	dominant frequency	0.277	0.360	–
16	Broadband	dominant frequency	0.322	0.306	–
17	0–10kHz	dominant frequency	0.322	0.304	✓
18	Broadband	margin factor	0.298	0.309	–
19	Broadband	crest factor	0.309	0.290	–
20	10–20kHz	kurtosis	0.232	0.347	–
21	0–10kHz	margin factor	0.260	0.289	–
22	0–10kHz	crest factor	0.269	0.253	–
23	Broadband	shape factor	0.238	0.280	–
24	20–100kHz	spectral bandwidth	0.267	0.245	✓
25	Broadband	skewness	0.298	0.206	✓
26	20–100kHz	spectral flatness	0.255	0.234	–
27	0–10kHz	shape factor	0.225	0.254	–
28	0–10kHz	skewness	0.276	0.190	–
29	10–20kHz	shape factor	0.113	0.305	–
30	20–100kHz	crest factor	0.184	0.230	–
31	20–100kHz	spectral skewness	0.263	0.144	✓
32	20–100kHz	center frequency	0.176	0.187	–
33	10–20kHz	spectral bandwidth	0.167	0.136	✓
34	10–20kHz	spectral flatness	0.164	0.112	–
35	10–20kHz	center frequency	0.158	0.108	–

CRediT authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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IFD?

Data availability

The reproducible Python analysis pipeline and extracted features (Parquet format) accompany this paper.

Public release of the complete dataset (HDF5 signal files) is not yet decided—resolve with CeramicSpeed and choose: (a) full Zenodo deposit, (b) features + code only, or (c) available from the corresponding author on reasonable request. Align contribution 4

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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